Windows Deployment Service

Windows Configuration Manager

Pxe_Booting eXecution Environment

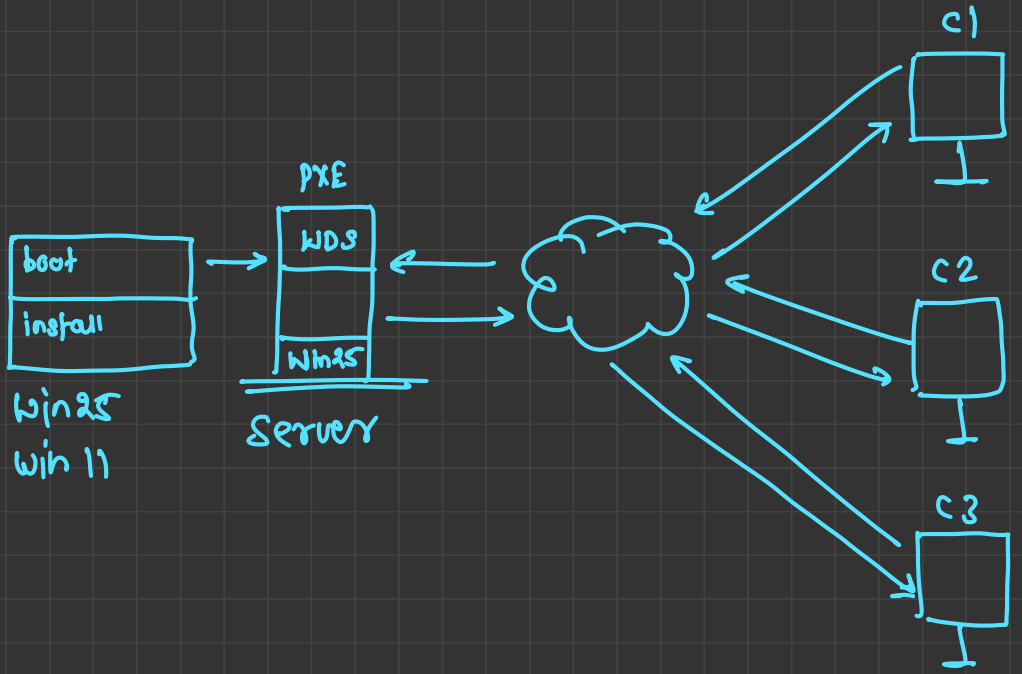
↳ network booting

Bios

↳ Boot order

↳ Network

↳ Disk



Introduction

- Windows Deployment Services (WDS) is a server role included with Windows Server that enables network-based installation of Windows operating systems
- It allows administrators to deploy Windows to computers that have no installed operating system by leveraging Preboot Execution Environment (PXE) boot technology
- Despite the deprecation announcements, the following WDS capabilities remain available on Windows Server 2025:
  - **PXE boot services:** WDS still functions as a PXE server, responding to DHCP requests from network boot clients
  - **Custom boot image deployment:** WDS can boot devices using custom Windows PE images (not the standard boot.wim from installation media)
  - **Image storage and management:** The WDS image store continues to manage boot and install images
  - **Multicast transmission:** The Transport Server component's multicast engine remains functional
  - **Deployment of older operating systems:** Windows 10, Windows Server 2019, and earlier versions can still be deployed using standard workflows

WDS Architecture

- **Transport Server**
  - The Transport Server provides the core networking functionality for WDS, specifically for multicast transmission
  - Its primary purpose is to distribute images efficiently to multiple computers simultaneously, conserving network bandwidth
- **Deployment Server**
  - The Deployment Server requires the Transport Server and provides the actual deployment orchestration
  - It includes the following sub-components :
    - **PXE Server:** Listens for DHCP requests and responds with PXE offers to network boot clients
    - **Image Store:** Manages boot images and install images uploaded to the WDS server
    - **WDS Client:** The component within Windows PE that connects to WDS, joins sessions, and applies install images
    - **TFTP Server:** Functions as a Trivial FTP server for boot file transfer
- **Image Types**
  - **Boot Image:**
    - Provides the Windows PE environment for deployment
    - Scaled-down Windows OS (Windows PE); typically one for 32-bit and one for 64-bit architectures
  - **Install Image:**
    - The actual operating system to be deployed
    - Mirror of a hard drive containing OS, applications, updates, roles, and settings

Installation

- **Prerequisites**
  - **DHCP Server:** Required to assign IP addresses to PXE clients (can be on the same or different server)
  - **Active Directory (for Integrated mode):** WDS can be configured in either AD-integrated or standalone mode
  - **DNS:** Proper name resolution for network services
  - **Dedicated storage:** A separate partition or disk for WDS image storage is strongly recommended
- **Installation Steps**
  - Open **Server Manager** and select **Add Roles and Features**
  - Choose **Role-based or feature-based installation**
  - Select the target server
  - Check **Windows Deployment Services**
  - When prompted, add the required features (management tools)
  - In the role services selection, both **Deployment Server** and **Transport Server** are selected by default—leave both enabled
  - Complete the wizard and install